EXERCISE 2.7

1. Write in polar form of the following complex numbers

(i) $2+i2\sqrt{3}$ (ii) $3-i\sqrt{3}$ (iii) $-2-i2$ (iv) $\dfrac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$.

2. Find the rectangular form of the complex numbers

(i) $\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)\left(\cos\frac{\pi}{12}+i\sin\frac{\pi}{12}\right)$ (ii) $\dfrac{\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}}{2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)}$.

3. If $(x_1+iy_1)(x_2+iy_2)(x_3+iy_3)\cdots(x_n+iy_n)=a+ib$, show that

(i) $\left(x_1^{\,2}+y_1^{\,2}\right)\left(x_2^{\,2}+y_2^{\,2}\right)\left(x_3^{\,2}+y_3^{\,2}\right)\cdots\left(x_n^{\,2}+y_n^{\,2}\right)=a^2+b^2$

(ii) $\sum_{r=1}^{n}\tan^{-1}\left(\frac{y_r}{x_r}\right)=\tan^{-1}\left(\frac{b}{a}\right)+2k\pi$, $k\in\mathbb{Z}$.

4. If $\dfrac{1+z}{1-z}=\cos 2\theta+i\sin 2\theta$, show that $z=i\tan\theta$.

5. If $\cos\alpha+\cos\beta+\cos\gamma=\sin\alpha+\sin\beta+\sin\gamma=0$, show that

(i) $\cos 3\alpha+\cos 3\beta+\cos 3\gamma=3\cos(\alpha+\beta+\gamma)$ and

(ii) $\sin 3\alpha+\sin 3\beta+\sin 3\gamma=3\sin(\alpha+\beta+\gamma)$.

6. If $z=x+iy$ and $\arg\left(\dfrac{z-i}{z+2}\right)=\dfrac{\pi}{4}$, show that $x^2+y^2+3x-3y+2=0$.

2.8 de Moivre's Theorem and its Applications



de Moivre
1667–1754

Abraham de Moivre (1667–1754) was one of the mathematicians to use complex numbers in trigonometry.

The formula $(\cos\theta+i\sin\theta)^n=(\cos n\theta+i\sin n\theta)$ known by his name, was instrumental in bringing trigonometry out of the realm of geometry and into that of analysis.

2.8.1 de Moivre's Theorem

de Moivre's Theorem

Given any complex number $\cos\theta+i\sin\theta$ and any integer n,

$$(\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta$$

Corollary

(1) $(\cos\theta-i\sin\theta)^n=\cos n\theta-i\sin n\theta$ (2) $(\cos\theta+i\sin\theta)^{-n}=\cos n\theta-i\sin n\theta$

(3) $(\cos\theta-i\sin\theta)^{-n}=\cos n\theta+i\sin n\theta$ (4) $\sin\theta+i\cos\theta=i(\cos\theta-i\sin\theta)$.

Now let us apply de Moivre's theorem to simplify complex numbers and to find solution of equations.

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